

PROTOTYPES & DEMOS

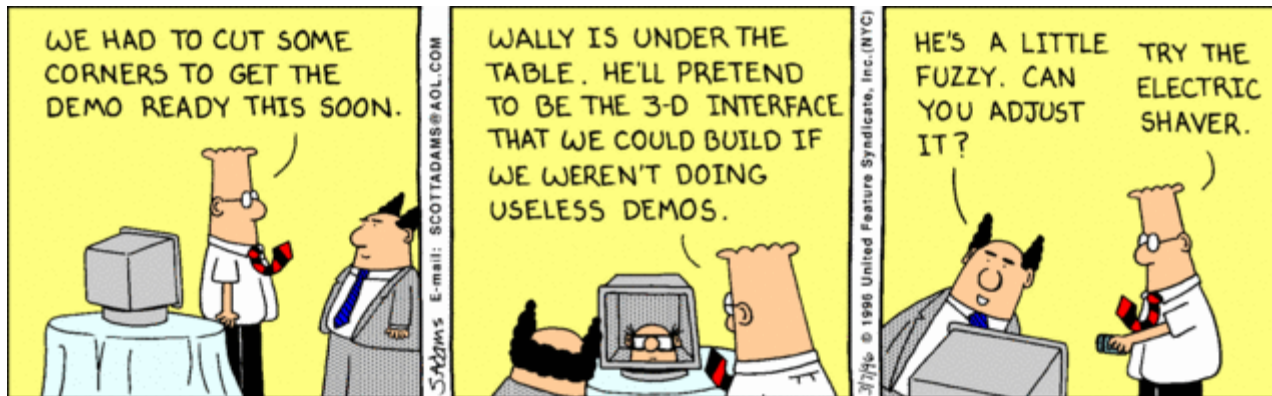
Edwin Blake
(with thanks to Gary Marsden and Scott Adams)



Brief Guidelines

2

- Highlight points to consider when prototyping
- Some ideas on how to conduct a software demonstration





3

Prototypes



What is a Prototype

4

- Here we only talk about program prototypes
 - ▣ Not, for example, paper prototypes.
- “Initial, incomplete version of a software system being developed that is used to learn about the problem, explore designs and solution techniques”
- Use it to investigate:
 - ▣ Technical issues
 - ▣ Work flow, task design
 - ▣ Screen layouts and information display
 - ▣ Difficult, controversial, critical areas
- These can only be very general guidelines as the actual prototype very much depends on the particular system you are working on.



Why Prototype?

5

- Evaluation and feedback are central to design
 - ▣ Stakeholders can see, hold, interact with a prototype more easily than a document or a drawing
 - ▣ Team members can communicate effectively
- You can test out ideas for yourself
 - ▣ It encourages reflection: very important aspect of design
- Prototypes answer questions, and support designers in choosing between alternatives



Compromises

6

- All prototypes involve compromises
- For software-based prototyping maybe there is a slow response? sketchy icons? limited functionality?
- Two common types of compromise
 - ▣ 'horizontal': provide a wide range of functions, but with little detail
 - often used to test user interface early
 - ▣ 'vertical': provide a lot of detail for only a few functions
- Compromises in prototypes mustn't be ignored.



Realising Prototypes: Throw-Away

7

- Address high risk issues
 - ▣ Uncertainty in requirements
 - ▣ Explore User interface design
 - ▣ Alternative implementation strategies
 - ▣ Technology platform
- Focus only on the issue
 - ▣ Build a system that addresses the problem and ignores all other concerns
- Throw-away (rapid prototyping): may suspend normal development standards
 - ▣ No unit tests
 - ▣ Done to try out alternative ideas with users



Realizing Prototypes: Evolutionary

8

- Quality must be attended to
- Can be put to *limited* use with customer
- May well form kernel of eventual system
- Can be used to implement well understood requirements and see how happy customer is with that
 - ▣ Add more as requirements are refined (a bit like agile anyway)



How does the prototype relate to the final system?

9

- **Throw-away (aka revolutionary)**
 - ▣ influences via requirements
- **Evolutionary (aka tracer-bullet)**
 - ▣ a part of the system is used to be the prototype
- **Iterative (Agile)**
 - ▣ phased implementation
 - ▣ 1 increment informs subsequent increments



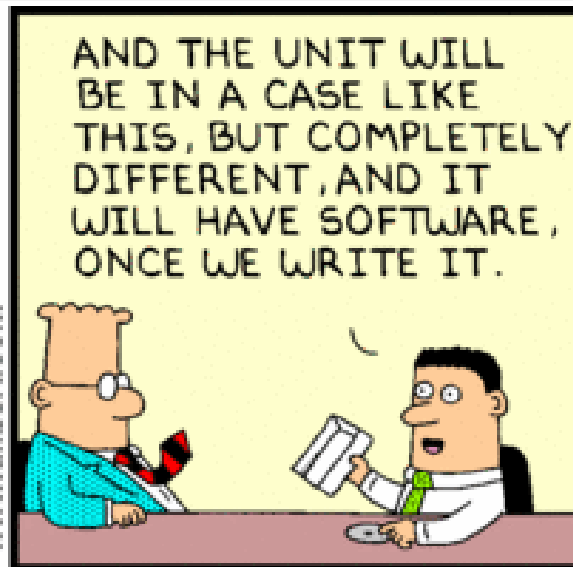


10

Demonstration



www.dilbert.com
scottadams@aol.com



2/4/03 © 2003 United Feature Syndicate, Inc.



Before the Demo



11

- **Make sure your system works**
- List the important tasks your demonstration system can do
 - Think through what you are going to say
 - Provide a very short overview of the system and explain what the demo will show
 - ... these three tasks are essential features to test ...
 - Our user interface is designed for ... and the following two scenarios are the use case we focussed on ...
 - Just as if you were giving a talk except here you *show* the issues — don't talk too much
 - Work out the steps involved in showing the tasks



Before the Demo



12

- Practice what you want to show
 - ▣ Ensure team members have an equal share in the demo and can demonstrate their contribution and knowledge
 - ▣ Ensure your demo fits into the time allowed
- Think of possible questions
 - ▣ Ensure that each person answers for themselves
 - ▣ Be ready to say what has been left out of the demo and why
 - ▣ Be ready to clarify roles of the team members



Just Before the Demo

13

- **Make sure your system works**
 - ▣ Test it on the actual system used for the demo
 - ▣ Resist any temptation to tweak the demo system
 - If you absolutely cannot resist “fixing” something
 - always keep the working version
 - be ready to demo the first version anyway
 - ▣ If you need special peripherals (e.g., headphones, webcams) ensure you bring them and test them too
- Arrive early
 - ▣ Be sure of the venue
 - ▣ If you need to demo on the same system as others
 - arrive well before anyone has to start
 - test that your demo runs on that actual system



During the Demo

14

- Stay calm
- Don't make effusive apologies if anything goes wrong
 - ▣ Fix problems quietly
 - ▣ Be able to restart the system without a recompilation
 - ▣ Let one person go on explaining the system
- Do not argue with or contradict team members
- Be prepared to change direction in response to questions
- Allow and encourage the client to try things (where safe to do so)
- Be prepared to show code or data or documentation





15

Any Questions?

